Unsupervised Denoising for Satellite Imagery Using Wavelet Directional CycleGAN

Joonyoung Song, Jae-Heon Jeong, Dae-Soon Park, Hyun-Ho Kim, Doo-Chun Seo, and Jong Chul Ye, *Fellow, IEEE*

***Abstract*— Multispectral satellite imaging sensors acquire various spectral band images and have a unique spectroscopic property in each band. Unfortunately, image artifacts from imaging sensor noise often affect the quality of scenes and have a negative impact on applications for satellite imagery. Recently, deep learning approaches have been extensively explored to remove noise in satellite imagery. Most deep learning denoising methods, however, follow a supervised learning scheme, which requires matched noisy image and clean image pairs that are difficult to collect in real situations. In this article, we propose a novel unsupervised multispectral denoising method for satellite imagery using a wavelet directional cycle-consistent adversarial network (WavCycleGAN). The proposed method is based on an unsupervised learning scheme using adversarial loss and cycle-consistency loss to overcome the lack of paired data. Moreover, in contrast to the standard image-domain cycleGAN, we introduce a wavelet directional learning scheme for effective denoising without sacrificing high-frequency components such as edges and detailed information. Experimental results for the removal of vertical stripes and wave noise in satellite imaging sensors demonstrate that the proposed method effectively removes noise and preserves important high-frequency features of satellite images.**

***Index Terms*— Cycle-consistent adversarial network (cycleGAN), image denoising, multispectral satellite imagery, unsupervised learning.**

I. Introduction

MULTISPECTRAL imaging sensors from a satellite capture different types of spectral band information. For instance, a typical high-resolution satellite has several imaging sensors for multispectral bands: red (R), green (G), blue (B), near-infrared (N), etc. Each spectral band signal has unique spectroscopic characteristics, resulting in a variety of remote sensing applications, such as agricultural planning [1], traffic monitoring [2], city planning [3], and disaster analysis [4].



(a) (b)

Fig. 1. Examples of satellite images with (a) vertical stripe noise and (b) wave noise.

Unfortunately, the quality of satellite images is often affected by various noise sources, such as system calibration errors, the intrinsic properties of the hardware, sensitivity of the sensors, the photon effect, and thermal noise. Fig. 1 shows typical examples of structured noise patterns in images from a high-resolution satellite, such as vertical stripe noise and wave noise. The main cause of vertical stripe noise is interference from the different scan timings of multispectral imaging sensors in a push broom scanner. Different sampling timings, and also the sensitivity of the sensors, induce a different offset in each detector and generate vertical stripe noise patterns. Horizontal wave noise is an irregular wave noise pattern that is caused by interference from various hardware components. Noise in images degrades the quality of the scenes and limit the use of satellite imagery. Therefore, one of the most important preprocessing steps for satellite images is the elimination of image noise that occurs during the image acquisition process.

Previously, various methods have been proposed for the removal of noise in satellite images. Conventional denoising methods follow model-based approaches using hand-crafted features and prior knowledge of data [5]–[16]. However, the limitation of traditional model-based methods is a degradation in performance if predefined features or prior knowledge of the model do not fully reflect the properties of new data.

Recently, deep convolutional neural networks (CNNs) have shown significant performance gain in image denoising

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Joonyoung Song and Jong Chul Ye are with the Department of Bio and Brain Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon 34141, South Korea (e-mail: songjy18@kaist.ac.kr; jong.ye@kaist.ac.kr).

Jae-Heon Jeong, Dae-Soon Park, Hyun-Ho Kim, and Doo-Chun Seo are with Image Data System Development Division, Satellite Information Center, Korea Aerospace Research Institute (KARI), Daejeon 34133, South Korea (e-mail: jjh583@kari.re.kr; parkds@kari.re.kr; kimhh@kari.re.kr; dcivil@kari.re.kr).

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problems [17]–[20]. The advantage of using deep learning methods comes from the data-driven nature that automatically learns the optimal features for the task from the data. In remote sensing applications, CNN-based denoising algorithms have also been proposed with promising results [21]–[27]. However, most CNN-based denoising methods for satellite images are trained in a supervised manner. Unfortunately, supervised learning schemes require structurally matched noisy image and clean target image pairs, which are difficult to obtain in real situations. To utilize unmatched image pairs, unsupervised learning methods should be used. Recently, unsupervised denoising methods using a deep image prior or self-consistency have been developed [28]–[30]. However, these methods are difficult to implement in the field of remote sensing due to their computational complexity or different assumptions about the models and data.

Another limitation of most CNN-based denoising algorithms for satellite imagery is that the methods are designed in the image domain, which often leads to blurred output and loss of information on the edges and details, especially when there is not sufficient training data. High-frequency components in satellite images contain important information that is crucial for the use of the satellite. Therefore, a desirable denoising algorithm should only remove noise components while preserving image details. Transform domain learning approaches are an alternative to image domain denoising methods. For instance, the advantage of using a wavelet transform is that the image can be decomposed to directional subbands that can be used effectively to remove noise while preserving high-frequency components. Accordingly, wavelet-based denoising methods have also been developed for remote sensing applications [31]–[33]. Moreover, by combining this with the CNN approach, a wavelet domain denoising method using deep learning was pioneered by our group for medical applications [34], which was followed by others for natural image and remote sensing [24], [35]. Unfortunately, these algorithms are still in a supervised learning framework which requires matched noisy and clean target image pairs.

Recently, unsupervised learning using a generative adversarial network (GAN) [36] has become increasingly popular. However, the standard GAN approaches often suffer from mode-collapsing behavior, which often generates artificial features. To mitigate the mode-collapsing problem, unsupervised image-to-image translation using the cycle-consistent adversarial network (CycleGAN) was proposed [37]. Inspired by the success of cycleGAN, our group [38] proposed an unsupervised denoising method using a cycle-consistent adversarial network for multiphase coronary computed tomography (CT) angiography. Here, the denoising problem was considered as an image-to-image translation problem between two domains: a noisy image domain and a clean image domain, and the results in [38] confirmed that cycleGAN is a promising tool for unsupervised denoising.

By synergistically combining these findings, we propose an unsupervised multispectral denoising method for satellite imagery using a wavelet directional cycle-consistent adversarial network (WavCycleGAN) and demonstrate its superior performance for the removal of two structured noise patterns: vertical stripe noise and wave noise. Specifically, based on the property of target noise, specific wavelet subbands that contain most of the noise are selected for the wavelet recomposition to obtain the wavelet directional image. Then, our cycleGAN network is trained in an unsupervised manner to learn the distribution matching between two wavelet directional images from clean and noisy domains. Our experimental results show that our multispectral denoising method using WavCycleGAN effectively removes vertical stripe noise and wave noise while preserving edges and details of images.

To the best of our knowledge, our WavCycleGAN is the first approach to use both an unsupervised learning scheme and wavelet domain processing for accurate denoising tasks. We confirmed our method with real-world satellite image artifact removal that requires a high precision algorithm.

Recall that a CNN is usually considered as image domain processing inspired by the information processing in the visual cortex, so most of the unsupervised learning tasks are performed in the image domain. Another important contribution of this work therefore is the first demonstration that unsupervised learning is feasible in the transform domain to effectively remove the specific directional artifact components.

In fact, this was inspired by our theoretical work where a cycleGAN can be derived from an optimal transport (OT) formulation that transports one probability measure to another by minimizing the average transport cost [39]. Hence, if the wavelet directional image distribution is well-structured and easier to transport, this work shows that it is better to use cycleGAN in that domain. Therefore, although our method was developed to remove noise in satellite images, it opens a new research direction for image denoising for various applications such as medical imaging, natural images, and other areas.

II. Related Works

A. Model Based Approaches

Conventional satellite image denoising methods typically exploit model-based approaches which utilize hand-crafted representation and the intrinsic properties of satellite images. For example, assuming that the satellite images tend to be piecewise smooth in the spatial domain [9], the total variation (TV) denoising model [40] has been applied for noise removal of satellite imagery because it effectively preserves high-frequency information and enforces piecewise smoothness [5], [6], [9]. Yuan *et al.* [5] extended the TV model to the spectral-spatial adaptive TV denoising model which considers spectral and spatial information of images. Chang *et al.* [6] proposed the image destriping method using the anisotropic spectral-spatial total variation model. He *et al.* [9] regularized their model with TV to enforce piecewise smoothness of images.

On the other hand, clean satellite images that consist of multispectral images can be considered to have low-rank properties [7]. Based on the intrinsic sparsity of satellite images, low-rank matrix recovery (LRMR) approaches have been applied to noise removal problems in remote sensing [7]–[9]. Zhang *et al.* [7] introduced an image restoration method using LRMR. He *et al.* [8] proposed noise-adjusted

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framework that takes into account different properties of noise in different bands. He *et al.* [9] introduced a total variation-regularized low-rank matrix factorization (LRTV). Recent studies have exploited tensor-based approaches with low-rank properties and TV regularization to utilize spatial-spectral correlations of multispectral satellite imagery [10]–[13].

Satellite imagery has various band images that contain highly redundant spectral-spatial information. Yuan *et al.* [14] and Zheng *et al.* [15] used the spectral-spatial relationship of satellite imagery to remove noise signals and obtained successful results. Lu *et al.* [16] proposed the sparse coding based method with a spike and slab prior to remove noise.

However, the drawback of these model-based image restoration methods is the use of hand-crafted features and data model, which may degrade the performance of the algorithm if the image data has unexpected properties beyond their assumptions. Moreover, the algorithms are usually computationally expensive.

B. Deep Learning Approaches

In the field of remote sensing, many deep learning based methods have been recently proposed for the removal of noise in satellite imagery. Yuan *et al.* [21] proposed a spatial-spectral deep residual learning method using a CNN for hyperspectral images (HSID-CNN). Their method utilized spatial and spectral information by using noisy input and adjacent spectral bands. Chang *et al.* [22] proposed a method for hyperspectral image denoising via a CNN (HSI-DeNet). HSI-DeNet consists of dilated convolution layers and exploits a residual learning approach to effectively remove noise. Zhang *et al.* [23] introduced a spatial-spectral gradient network (SSGN) for the removal of hybrid noise in hyperspectral images. The SSGN uses spatial and spectral gradient information to extract important features of satellite images. Guan *et al.* [24] proposed a wavelet deep neural network for stripe noise removal. They trained the network in the wavelet domain to achieve effective denoising. Other deep learning based denoising methods for satellite imagery can be found in [25], [26], and [27].

Most deep learning based image restorations for satellite imagery follow a supervised learning scheme. A supervised learning method requires a paired data set consisting of a noisy image and spatially matched clean image to train the network. The need for a paired data set, however, limits the use of a supervised learning scheme in practice, because it is difficult to collect paired satellite images in real situations. To mitigate this problem, many researchers added synthesized noise to relatively clean images. Although the trained model using synthetic noise works well for artificial noise, it is difficult to estimate real noise components that are complex in practice.

In contrast to a supervised learning approach, an unsupervised denoising framework does not need matched noisy image and clean target image pairs. Unsupervised denoising methods have been proposed for denoising problems [28]–[30]. Ulyanov *et al.* [28] proposed the deep image prior (DIP) approach and showed that a deep neural network itself can be used as an image prior, which leads to an appealing denoising performance using only noisy images. However, DIP is computationally very expensive and even higher than the model-based approach, which prohibits its use for satellite imagery. Lehtinen *et al.* [29] introduced the Noise2Noise approach to solve image restoration problems without clean images. However, Noise2Noise needs image pairs consisting of different noisy realizations from the same clean image, which is not possible for remote sensing applications. Krull *et al.* [30] introduced a Noise2Void framework in which the network learns how to remove noise from single noisy images. However, Noise2Void assumes that the noise is pixel-wise independent, which was not the case for stripe and wave noise in this study.

One of the recent powerful tools for unsupervised learning approaches is the GAN [36]. In a GAN framework, two neural networks are used: a generator and a discriminator. The goal of the generator in a GAN is to convert input to output that follows a target domain distribution. In contrast, the goal of the discriminator in a GAN is to determine whether sample data comes from the model distribution or the target domain distribution. To achieve this, an adversarial loss is used to train the generator and the discriminator. However, GAN has a problem of mode collapse that creates artifacts [37]. To alleviate the problem of mode collapse in the GAN, the CycleGAN was developed using an adversarial loss and a cycle-consistency loss [37]. Inspired by the CycleGAN, our group [38] first proposed the cycle-consistent adversarial denoising network for multiphase coronary CT imaging in the field of medical image reconstruction and showed appealing results. In our recent follow-up work [39], the CycleGAN was shown to be related to the OT problem, where the goal is to transport the probability measure to another at minimal cost. Therefore, a properly designed cycleGAN can be safely used for image denoising tasks without being concerned about creating artificial features. However, as these CycleGANs are usually designed in the image domain, this often degrades the quality of high-frequency image details that are critical to applications of satellite imagery.

C. Wavelet Based Approaches

The wavelet transform is useful to decompose a signal into different subbands at multiple directions and scales, which makes it possible to design effective denoising methods. Previously, wavelet-based methods were proposed to remove noise in satellite imagery [31]–[33]. Chen *et al.* [31] proposed a destriping method using a discrete wavelet transform (DWT) to capture the frequency components of stripes. Chen *et al.* [32] developed the denoising technique using a principal component analysis (PCA) and wavelet shrinkage approach. Pande-Chhetri *et al.* [33] proposed a denoising method using a wavelet transform with an adaptive frequency domain filtering approach. The wavelet transform can be combined with deep learning based methods in image reconstruction problems as in [24], [34], and [35]. Initially for medical imaging applications, our group [34] first introduced a deep learning method based on a wavelet transform to remove noise in low-dose CT images. Inspired by this success,

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Liu *et al.* [35] proposed a multilevel wavelet-CNN model for image restoration tasks. Guan *et al.* [24] developed a wavelet deep neural network to remove stripes. However, these methods follow a supervised learning scheme which requires structurally paired images, which are difficult to collect.

Recently, wavelet-based GAN methods were proposed for face hallucination [41] and face aging [42]. Huang *et al.* [41] proposed a wavelet domain GAN for face hallucination to reconstruct both local details and global features of face images. However, the framework in [41] requires a spatially paired data set. Liu *et al.* [42] used a wavelet transform to improve the visual quality of results based on the fact that age-related features can be captured by a wavelet transform. However, the method in [42] reconstructs image domain data, which often leads to loss of image details.

D. Our Contributions

In contrast, our method follows an unsupervised learning scheme which does not require a paired data set. Furthermore, our WavCycleGAN removes noise patterns in the wavelet directional image without sacrificing high-frequency details. To the best of our knowledge, our WavCycleGAN is the first approach to exploit the best of the two methods. Our contributions can be summarized as follows.

1) Unsupervised Learning Approach: Although it is difficult to obtain matched clean and noisy image pairs from satellite imagery, it is much easier to obtain *unmatched* clean and noisy image data sets in practice. This is because in some situations sensors are affected less by the noise, or the assumption of the model-based approaches are sufficiently accurate to generate clean images. However, the practical issue is that these clean images are not matched to the noisy multispectral image data that one is interested in processing. In this scenario, an unsupervised learning scheme that uses the unmatched clean and noisy image data set is a perfect fit.

Therefore, one of the most important contributions of our article is an unsupervised learning scheme that uses the unmatched clean and noisy images for neural network training.

2) Wavelet Directional Learning: Unlike for the existing deep learning based algorithms designed in the image domain, we train our model using wavelet directional images that are obtained from a subset of wavelet bands containing noise. The motivation of using WavCycleGAN is that if noise patterns are structured in a directional manner, directional components of the noise can make up the most of the noise patterns. As a result, the spectral contents in the other bands are not altered by the neural network so that we can achieve efficient noise removal without sacrificing high-frequency information.

III. Theory

A. Optimal Transport Driven CycleGAN

Here, we provide a brief review of our recent theoretical work [39] showing that the cycleGAN can be derived as an unsupervised way of distribution matching using the OT theory [43], [44]. The theory guarantees that properly designed and trained cycleGAN image denoising does not incur cosmetic changes, rather it provides real image improvement [39].

Specifically, let $\mathcal{X}$ and $\mathcal{Y}$ are noiseless and noisy signal domains, respectively. Formally, we say that $T : \mathcal{X} \mapsto \mathcal{Y}$ transports the probability measure $\mu \in P(\mathcal{X})$ to another measure $\nu \in P(\mathcal{Y})$, if

$$\nu(B) = \mu(T^{-1}(B)), \quad \text{for all } \nu\text{-measurable sets } B. \tag{1}$$

Suppose there is a cost function $c : \mathcal{X} \times \mathcal{Y} \to \mathbb{R} \cup \{\infty\}$ such that $c(x, y)$ represents the cost of moving one unit of mass from $x \in \mathcal{X}$ to $y \in \mathcal{Y}$. Monge's original OT problem [43], [44] is then to find a transport map T that transports μ to ν at the minimum total transportation cost. Furthermore, further simplification can be obtained from Kantorovich's dual OT formulation as follows:

$$\min_{\pi \in \Pi(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{Y}} c(x, y) d\pi(x, y) = \sup_{\varphi \in L^1(\mu)} \left\{ \int_{\mathcal{X}} \varphi(x) d\mu(x) + \int_{\mathcal{Y}} \varphi^c(y) d\nu(y) \right\} \tag{2}$$

where $\varphi^c(y)$ is the so-called c-transform of the Kantorovich potential $\varphi(y)$ and $\Pi(\mu, \nu)$ is a joint distribution whose marginal distribution matches μ and ν, respectively, and the above maximum is taken over the so-called Kantorovich potentials φ.

In our prior work [39], we proposed a new penalized least square (PLS) cost function with a novel deep learning prior

$$c(x, y; \Theta, \Upsilon) = \|y - F_\Lambda(x)\| + \|G_\Theta(y) - x\| \tag{3}$$

where G_Θ is a neural network with parameter Θ to estimate the unknown noiseless signal from noisy signal y, and $F_\Lambda(x)$ is a noisy image generating process from noiseless signal x that is described by a neural network with parameter Λ. One of the most important observations in [39] was that if (3) is used as a transport cost in OT formulation (2), then a cycleGAN architecture emerges. Therefore, the cycleGAN can be considered as a distribution matching process, so if properly designed and trained, the denoising process is real and we do not need to be concerned about the artificial feature generation.

B. Wavelet Directional Cycle-Consistent Adversarial Network

Given that the cycleGAN is a distribution matching process, the learning procedure becomes easier if the appropriate signal domain is found where learnable features are easy to see. For the stripes and wave artifacts in satellite images, we found that the wavelet domain is an appropriate domain for learning. More details are provided below.

1) Wavelet Directional Image: The procedure for generating wavelet directional images is as follows. First, we use the 2-D Daubechies-3 wavelet transform (db3) to decompose the input image to subband images, such as approximation (LL), horizontal detail (LH), vertical detail (HL), and diagonal detail (HH) bands. With the Kth level wavelet decomposition, we have $\{\mathrm{LL}_i\}_{i=1}^K$, $\{\mathrm{LH}_i\}_{i=1}^K$, $\{\mathrm{HL}_i\}_{i=1}^K$, and $\{\mathrm{HH}_i\}_{i=1}^K$ subband images. The advantage of using the wavelet transform is that we can decompose an image into directional subbands. Therefore, if the noise has specific directional properties, it can usually be confined in a specific subset of wavelet bands. This is the prior information we want to explore in designing the neural network.

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The wavelet decomposition level is determined based on the characteristics of the target noise. Specifically, the wavelet decomposition level is determined when most of the noise patterns only exist in detail bands. We use the DWT because it is simple to implement and provides a sparse representation of signals, while the continuous wavelet transform (CWT) and the stationary wavelet transform (SWT) generate a redundant representation of data and require more computation than the DWT.

Second, to generate wavelet directional images, wavelet recomposition is implemented only with selected wavelet subbands based on the directional property of target noise patterns. For example, for the case of vertical stripe noise in Fig. 2(a), the wavelet directional images are obtained by wavelet recomposition using the vertical detail subbands $\{\mathrm{HL}_i\}_{i=1}^{9}$ and zeroing out the other bands. This generates a wavelet directional image with the same dimensions as the input image as shown in Fig. 2(b), which clearly shows the noise signals without much of the underlying structures of the scene. In the case of images corrupted with wave noise as shown in Fig. 2(c), we find that the subbands $\{\mathrm{LH}_i\}_{i=1}^{6}$ contain most of the noise, so we use these bands to obtain the wavelet directional images in Fig. 2(d). The benefit of a wavelet directional image is that its dimensions are the same as those of the input image regardless of its level of decomposition. Note that the dimensions of each subband wavelet coefficient vary depending on the decomposition level, so it is difficult to implement the use of wavelet coefficients as direct input for the deep networks.

After a denoising network removes noise patterns in noisy wavelet directional images, clean output images can be acquired by subtracting predicted noise patterns from the noisy images.

2) Loss Formulation: We consider two domains: a clean domain ($\mathcal{X}$) and a noisy domain ($\mathcal{Y}$). The clean domain contains wavelet directional images without noise patterns, while the noisy domain consists of wavelet directional images with noise patterns. The two data domains are composed of data that are not matched to each other. $P_{\mathcal{X}}$ and $P_{\mathcal{Y}}$ are probability distributions of the clean domain and noisy domain, respectively. y is a sample from the noisy wavelet directional image distribution, and x is a sample from the clean wavelet directional image distribution. As shown in Fig. 3, $G_\Theta : \mathcal{Y} \mapsto \mathcal{X}$ is the generator parameterized with Θ, which converts a noisy wavelet directional image to a clean wavelet directional image; the generator $F_\Lambda : \mathcal{X} \mapsto \mathcal{Y}$ is a generator parameterized by Λ which generates a synthetic noisy wavelet directional image from a clean wavelet directional image. ψ_Ξ is an adversarial discriminator parameterized by Ξ that distinguishes synthetic noisy wavelet directionals from real noisy wavelet directionals. Similarly, φ_Φ is adversarial discriminators that distinguish denoised wavelet directional images from real clean wavelet directional images.

To train the WavCycleGAN for the denoising problem, our overall loss function consists of three loss functions: adversarial loss ℓ_{GAN}, cycle-consistency loss ℓ_{cycle}, and identity loss ℓ_{identity}. Adversarial loss is used to train the generators G_Θ and F_Λ, which learn the distribution matching in different domains. Cycle-consistency loss serves to impose one-to-one mapping between the input and the output on the generators G_Θ and F_Λ to maintain important information of the input image. Identity loss is used to make the generators G_Θ and F_Λ act as an identity function for the target domain image so that the output of a generator retains the input image which is from the target domain. Specifically, the typical adversarial loss for the generator G_Θ and the discriminator φ_Φ is as follows:

$$\ell_{\mathrm{GAN}}(\Theta, \Phi) = \mathbb{E}_{x\sim P_{\mathcal{X}}}[\log \varphi_\Phi(x)] + \mathbb{E}_{y\sim P_{\mathcal{Y}}}[\log(1 - \varphi_\Phi(G_\Theta(y)))]. \quad (4)$$

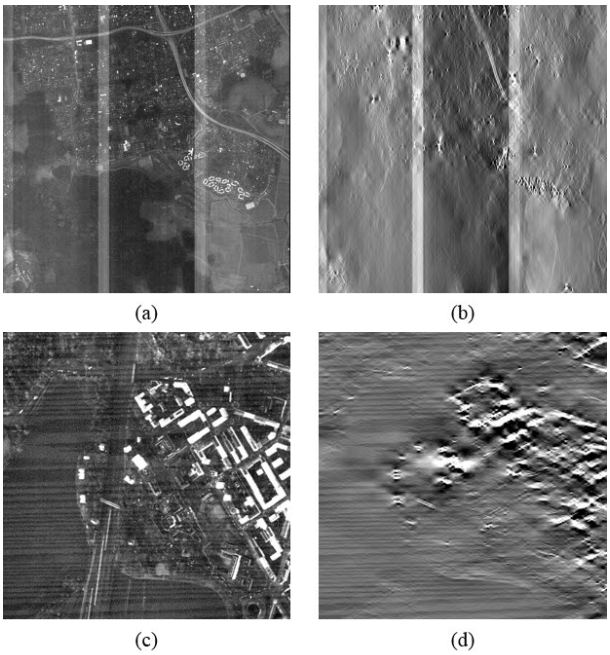


Fig. 2. Examples of satellite images and wavelet directional images. (a) Image with vertical stripe noise, (b) vertical wavelet directional image from (a). (c) Image with wave noise, and (d) horizontal wavelet directional image from (c).

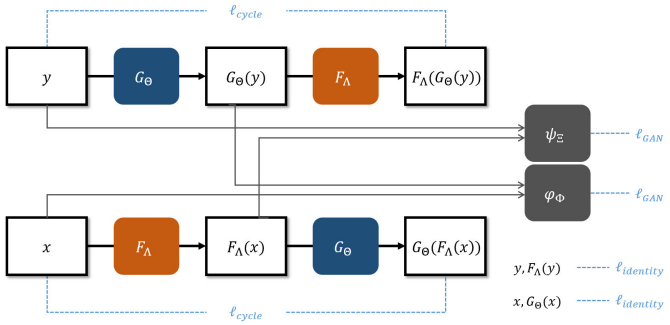


Fig. 3. Architecture of our WavCycleGAN for denoising satellite images. x and y are wavelet directional images from the clean domain $\mathcal{X}$ and the noisy domain $\mathcal{Y}$, respectively. The full objective consists of the adversarial loss ℓ_{GAN}, cycle-consistency loss ℓ_{cycle}, and identity loss ℓ_{identity}. By optimizing the full objective function with respect to generators (G_Θ and F_Λ) and discriminators (ψ_Ξ and φ_Φ), the denoising network G_Θ can be trained in an unsupervised manner using wavelet directional images.

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To train G_Θ and φ_Φ, we need to solve the min–max problem as follows:

$$\min_{\Theta} \max_{\Phi} \ \ell_{\text{GAN}}(\Theta, \Phi). \tag{5}$$

The least squares GAN (LSGAN) [45] uses the least square loss function instead of the cross entropy loss to overcome the problem of vanishing gradients. Therefore, rather than using the original GAN in (4), we adopted the LSGAN for the min-max problem as follows:

$$\min_{\Theta} \ \mathbb{E}_{y\sim P_{\mathcal{Y}}}[(\varphi_\Phi(G_\Theta(y)) - 1)^2] \tag{6}$$

$$\min_{\Phi} \ \frac{1}{2}\mathbb{E}_{x\sim P_{\mathcal{X}}}[(\varphi_\Phi(x) - 1)^2] + \frac{1}{2}\mathbb{E}_{y\sim P_{\mathcal{Y}}}[\varphi_\Phi(G_\Theta(y))^2]. \tag{7}$$

By solving the min–max game, G_Θ is trained to generate synthesized clean wavelet directional images from real noisy wavelet directional images and deceive the discriminator φ_Φ, while φ_Φ learns to discriminate between synthesized clean wavelet directional images $G_\Theta(y)$ and real clean wavelet directional images x. When the networks converge, G_Θ produces realistic clean wavelet directional images, and φ_Φ cannot distinguish between real clean wavelet directional images and synthesized clean wavelet directional images from G_Θ. The role of the adversarial loss for F_Λ and ψ_Ξ is similar to that of G_Θ and φ_Φ.

However, using only the adversarial loss may produce artificial features due to the mode collapsing problem. We used cycle-consistency loss to impose one-to-one mapping between input images and output images to reduce artifacts and to maintain important features other than noise components. The cycle-consistency loss is defined using the L1 norm as follows:

$$\ell_{\text{cycle}}(\Theta, \Lambda) = \mathbb{E}_{y\sim P_{\mathcal{Y}}}[||F_\Lambda(G_\Theta(y)) - y||_1] + \mathbb{E}_{x\sim P_{\mathcal{X}}}[||G_\Theta(F_\Lambda(x)) - x||_1]. \tag{8}$$

By enforcing the cycle-consistency for the networks, the generators G_Θ and F_Λ can be inverse mappings of each other, in which important features of images can be maintained during the domain translation.

Once the network is trained, at the inference phase, only the denoiser G_Θ is used. However, in many practical situations, many input images or image patches for the denoiser G_Θ may not be corrupted by the noise patterns. A desired generator G_Θ therefore should remove the noise pattern in the noisy wavelet directional image while maintaining the input wavelet directional images if there is no noise. Moreover, the desired generator F_Λ adds the noise pattern when the input is clean, while it maintains the input image when the input has the noise pattern. This condition is often called the identity property, i.e., $G_\Theta(x) \simeq x$ and $F_\Lambda(y) \simeq y$ [38]. In the original work on the CycleGAN [37], identity loss was used to maintain the color composition of the input, so a generator does not change the tone of the input image. In our work, however, we exploit the identity property to make a generator that preserves the input image from the target domain. To enforce the identity property, we define the identity loss using the L1 norm as follows:

$$\ell_{\text{identity}}(\Theta, \Lambda) = \mathbb{E}_{x\sim P_{\mathcal{X}}}[||G_\Theta(x) - x||_1] + \mathbb{E}_{y\sim P_{\mathcal{Y}}}[||F_\Lambda(y) - y||_1]. \tag{9}$$

To train the network by the adversarial learning approach with cycle-consistency and identity property, the overall loss function is defined using ℓ_{GAN} with the LSGAN scheme (6), (7), ℓ_{cycle} (8), and ℓ_{identity} (9) as follows:

$$\ell(G_\Theta, F_\Lambda, \psi_\Xi, \varphi_\Phi) = \ell_{\text{GAN}}(\Theta, \Phi) + \ell_{\text{GAN}}(\Lambda, \Xi) + \lambda\ell_{\text{cycle}}(\Theta, \Lambda) + \gamma\,\ell_{\text{identity}}(\Theta, \Lambda) \tag{10}$$

where λ and γ are hyperparameters for controlling the ratio of the losses between ℓ_{GAN}, ℓ_{cycle}, and ℓ_{identity}. To train the WavCycleGAN for the denoising problem, we aim to optimize the following the problem:

$$\min_{\Theta,\Lambda} \max_{\Xi,\Phi} \ \ell(G_\Theta, F_\Lambda, \psi_\Xi, \varphi_\Phi). \tag{11}$$

The corresponding architecture is given in Fig. 3. Notice that our WavCycleGAN uses wavelet directional images consisting of selected directional subbands, while the standard CycleGAN uses typical images. By considering the prior knowledge of noise patterns, the networks easily learn the properties of structured noise patterns and achieve improved performance compared to results of learning typical images, as we explain in the experimental section.

C. Reconstruction Flow for Specific Noise Patterns

1) Vertical Stripe Noise Removal: We found that vertical stripe noise patterns are distributed globally in images. Therefore, the networks need to see the image in full resolution and capture the overall trend of the stripe patterns to learn the relationship between clean and noisy images. However, the full resolution of a test scene with vertical stripe noise patterns is 3000×3000 pixels, which requires huge GPU memory and high computational cost. Moreover, the real training data are small, so we need a way to enrich the data set. To address problems, we used the prior knowledge that vertical stripe noise patterns are similar in the vertical direction.

Specifically, the following procedure is performed. First, our wavelet directional image is obtained by wavelet decomposition, noisy band selection, and wavelet recomposition. This makes the resulting wavelet directional image have the same size as the original images. As this is too big to upload in GPU memory for CNN training, additional downsampling is performed to reduce the size of the wavelet directional images. Next, to increase the training data size, we use random patch cropping to extract multiple patches from the downsampled wavelet directional images.

More specifically, we made the wavelet directional images using the vertical detail subbands $\{\text{HL}_i\}_{i=1}^9$ and applied downsampling along the vertical direction of the wavelet directional images by a factor of 32. We used the linear spline interpolation and Gaussian smoothing for downsampling to avoid aliasing. By using downsampled wavelet directional images, the networks can be trained using images with the global appearance of the stripe pattern and the computational costs can also be reduced. To train the WavCycleGAN for the removal of vertical stripe noise, we used randomly cropped patches with a size of 64×2048 pixels from downsampled

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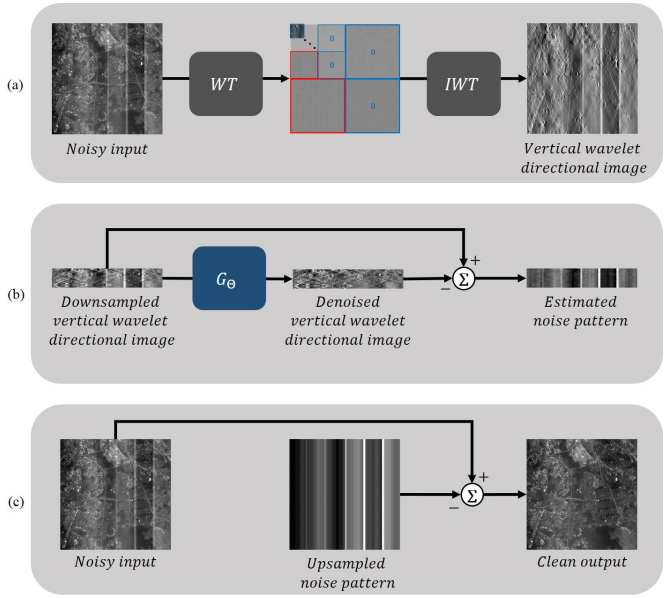


Fig. 4. Overall flow of our denoising method for the vertical stripe noise. (a) Process of making a vertical wavelet directional image. WT denotes a wavelet transform, and IWT refers to an inverse wavelet transform. The red-colored subbands are only used for wavelet recomposition. (b) Process of estimating a noise pattern. (c) Process of reconstructing a clean output from a noisy input and an upsampled noise pattern.

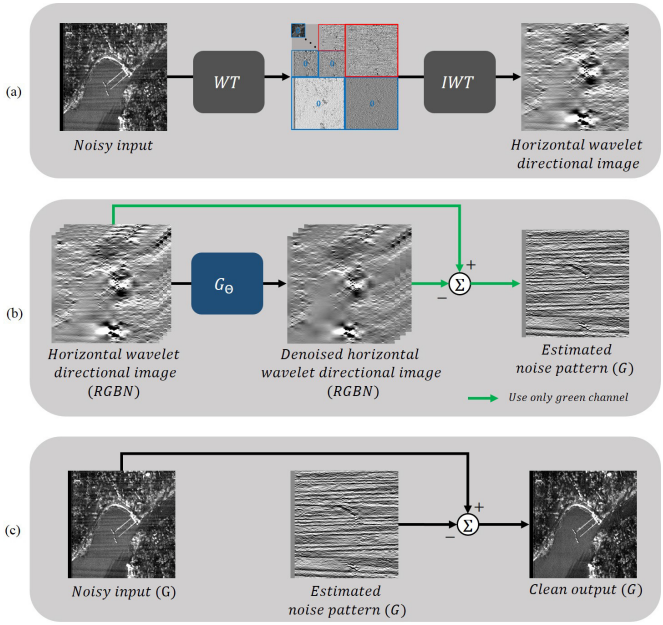


Fig. 5. Overall framework of our denoising method for the wave noise. (a) Procedure for generating a horizontal wavelet directional image. WT and IWT denote a wavelet transform and an inverse wavelet transform, respectively. We created a horizontal wavelet directional image for each spectral-band image. The red-colored subbands are only used for wavelet recomposition. (b) Process of estimating a noise pattern. The predicted noise pattern can be calculated using wavelet directional images of the green band. (c) Process of reconstructing a clean output image from a noisy image and a predicted noise pattern.

vertical wavelet directional images that show a wider view of the scenes than patches from original wavelet directional images. Thanks to the random cropping of patches, the amount of training data can be increased.

Fig. 4 shows the overall flow of our denoising method for the vertical stripe noise. First, the vertical wavelet directional image is generated using a db3 wavelet transform at the 9 decomposition level from the noisy input. When we apply the inverse wavelet transform, we preserve the coefficients of vertical bands (HL bands) and set the coefficients of the other bands (LL, LH, and LL bands) to zero. Second, the generator G_Θ removes the noise pattern of the downsampled vertical wavelet directional image. The estimated noise pattern is acquired by subtracting the denoised wavelet directional image from the downsampled wavelet directional image. Finally, the clean output is reconstructed by subtracting the upsampled noise pattern from the noisy input.

After the training process, we can only use the denoiser G_Θ in the inference stage for the denoising problem. Specifically, the noise pattern is calculated by subtracting the wavelet directional image reconstructed by the generator G_Θ from the noisy wavelet directional image. However, due to the downsampling of the input image by a factor of 32, the resolution of the estimated noise pattern differs from that of the input image. To increase the resolution of the noise pattern, we applied the upsampling process to the estimated noise pattern. The final reconstruction result is obtained by subtracting the upsampled noise pattern from the noisy input.

2) Horizontal Wave Noise Removal: We utilized spatially registered RGBN images for the removal of wave noise. The use of spatially registered RGBN images makes the network use the spatial correlation between multichannel images, which improves the reconstruction performance. To train the WavCycleGAN for the removal of wave noise, we used randomly cropped RGBN image patches with a size of 128 × 128 pixels from horizontal wavelet directional images.

Fig. 5 shows the overall framework of our denoising method for the wave noise. The first step is generating horizontal wavelet directional images. We produced horizontal wavelet directional images for each channel image. Next, the reconstructed horizontal wavelet directional image can be acquired using the generator G_Θ. In the data that we used, only green channel images had a wave noise pattern due to the specific sensor type. Therefore, we calculated the noise pattern by subtracting the predicted horizontal wavelet directional image from the noisy horizontal wavelet directional image of the green channel. Finally, the clean output image was acquired by subtracting the estimated noise pattern from the noisy input image of the green channel.

At the inference stage, the algorithms are applied by overlapping patch images by half to avoid blocking artifacts. The reconstructed full scene is then acquired by assembling only center parts of reconstructed patches. Specifically, we reconstruct 128 × 128 pixel patches and use only the center parts of patches with a size of 64 × 64 pixels.

IV. Methods

A. Data Set

1) Real Noisy Data: In this study, we utilized multispectral images from a high-resolution satellite. The multispectral images were composed of multispectral images from red (R), green (G), blue (B), near-infrared (N) imaging sensors. The

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data set was corrupted by either stripe noise or wave noise depending on the type of satellite imagery.

In order to develop a denoising algorithm for the vertical stripe noise that was mainly contained in the B channel, we used blue channel images from 14 scenes with a size of 6000 × 3000 pixels. This was because the multispectral data were provided by the data distributor (Korea Aerospace Research Institute: KARI) without registration, so we could not use them all together. For the removal of wave noise, we used 16 RGBN images in which each band image had a size of 6000 × 6000 pixels. In this case, the RGBN data were distributed by KARI with the image registration, so we were able to exploit the multispectral band redundancy. Horizontal wave noise was caused by hardware interference. In our data, which were distributed by Korea Aerospace Research Institute (KARI), only green channel images had wave noise patterns, which suggested that only the image sensor for the green band was affected by interference from hardware components that created wave noise.

For every scene, the upper part was used as training data and the lower part was used as test data. In a real situation, it is difficult to have completely clean images. To get clean images, we applied the conventional model-based reconstruction methods and used the resulting processed images as clean image references for training the denoising network in unsupervised setups.

2) Synthetic Noisy Data: The development of denoising algorithms with real samples creates the difficulty of quantitative evaluation, as there is no clean ground truth corresponding to a noisy image. Without ground truths, it is not possible to calculate quantitative metrics for the image reconstruction such as the peak signal-to-noise ratio (PSNR) and structural similarity index metric (SSIM) [46]. For quantitative evaluation of the algorithm, we therefore added synthesized noise to relatively clean data. To obtain a ground truth image for quantitative evaluation, we obtained synthetic noise patterns by subtracting the conventional model-based reconstruction results from the noisy images. Then, the synthetic noise patterns were added to the ground truth image to generate synthetic noisy image data.

For the task of vertical stripe noise removal, we generated eight synthetic image pairs with a size of 3000 × 3000 pixels that were not used for the training of the denoising network. For the wave noise removal, we generated four synthetic image pairs with a size of 3000 × 6000 pixels which were never used in the training data. When we added synthesized noise to green channel images, spatial correlation with other channels (red, blue, and near-infrared bands) was different from that with real data. For instance, if the pixel values of the synthesized noisy image exceeded the specified interval (e.g. [0, 65535]), values outside the interval needed to be clipped, which created an incorrect spatial correlation with other bands. Therefore, for a quantitative evaluation of horizontal wave images, we only compared results of the neural network using green band images. It was remarkable that these synthetic data were only used at the inference phase.

3) External Data: To examine the generalization ability of our model to other image data sets, we used the UC Merced (UCM) data set [47]. With the UCM data set, we trained and tested our framework for stripe noise removal and wave noise removal. The UCM data set consists of images of 21 land-use classes that contain extensive image components such as airplanes, buildings, roads, and forests. Each class consists of 100 images with a size of 256 × 256 pixels from aerial orthoimagery with a pixel resolution of one foot. We randomly shuffled 2100 images and 1680 images were used as the training set, while the other images were the test set. Similar to the simulation study with synthetic noisy data, we only used green band images. We created noisy images by synthesizing noise patterns in clean images. Specifically, the intensities of stripes were sampled from a normal distribution with zero mean and a standard deviation of 25. By adding these to wavelet vertical bands $\{\mathrm{HL}_i\}_{i=1}^{2}$, we generated noise images. To synthesize horizontally structured noise, we convolved Gaussian noise with the specified kernel as in [48]. Specifically, we used Gaussian noise (mean 0 and standard deviation 50) and the bar-shaped kernel with a size of 1 × 10 pixels. We added that noise in each specific wavelet horizontal band $\{\mathrm{LH}_i\}_{i=1}^{2}$. The resulting noise patterns were added to generate synthesized noisy images.

B. Implementation Details

1) Architecture of Generators and Discriminators: For the generators G_Θ and F_Λ in our denoising model, we used a tight-frame U-net [49] structure with a skip connection between the input and output nodes. The tight-frame U-net uses wavelet decomposition and concatenation instead of conventional pooling and unpooling layers in order to satisfy the frame condition so that the networks effectively reconstruct high-frequency components [50]. Furthermore, by adding the skip-connection between input and output nodes, we exploited the residual learning scheme, which was effective for denoising [19]. We also replaced batch normalization layers [51] with instance normalization layers [52] which is known for improving the quality of image generation. The discriminators φ_Φ and ψ_Ξ are constructed based on the structure of PatchGAN [53], which penalizes image patches to capture the texture and style of images. We used PatchGAN consisting of five convolutional layers and the fully connected layer with the instance normalization.

2) Training Details (Real Noisy Data): For the success of deep neural networks, a large amount of training data is often required. In addition, the variety of samples is an important factor. However, in many situations, a large number of data sets are not available, and we considered such an extreme situation to validate the advantages of our network.

Specifically, due to security issues, our training data had fewer than 20 scenes for the development of the noise removal algorithm. To mitigate the deficiency of training data, we utilized image patches cropped from the full scenes. Specifically, we randomly cropped image patches with a size of 64 × 2048 pixels from the downsampled vertical wavelet directional images for the algorithm of denoising the vertical stripe noise. For the denoising method of the wave noise, we utilized image patches with a size of 128 × 128 pixels randomly

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cropped from the horizontal wavelet directional images. We also used data augmentation strategies, such as horizontal flipping and vertical flipping. The use of image patches, which were cropped randomly at each iteration in the training phase, increased the variety of the samples, and a large number of training images were acquired.

For unsupervised training, we randomly shuffled image pairs so that the network used unmatched data for the training. Our WavCycleGAN was trained by solving the optimization problem (11) with $\lambda = 10$ and $\gamma = 5$. The size of mini-batch was 1. An Adam optimizer [54] was used to optimize the loss function with $\beta_1 = 0.5$ and $\beta_2 = 0.999$. The network was trained for 200 epochs. The initial learning rate was 2×10^{-3} during the first 100 epochs, and gradually decreased to 0 through the last 100 epochs. The implementation of our method was based on PyTorch library [55] using a NVIDIA GeForce GTX 1080 Ti GPU.

3) Training Details (External Data): For the stripe noise removal, we created a wavelet directional image by selecting vertical bands in decomposition level 2 and applied downsampling to the wavelet directional image by a factor of 2. To train our model, we used randomly cropped patches with a size of 64×256 pixels from downsampled wavelet directional images. To train the networks for the task of horizontally structured noise removal, we used wavelet directional images that were generated with the horizontal bands in decomposition level 2. We also used randomly sampled patches with a size of 128×128 pixels from wavelet directional images to train the networks. The other settings were the same as the settings of the experiment with real noisy data.

C. Comparative Methods

To evaluate the performance of the vertical stripe noise removal, we compared our method (WavCycleGAN) with various methods. Specifically, in order to investigate the effectiveness of learning wavelet directional images, we generated reconstruction results using a standard image domain cycleGAN (CycleGAN) that does not utilize any directional decomposition using a wavelet transform. Furthermore, we generated results from the GAN model designed by removing cycle consistency and using only a pair of a generator and discriminator. We also compared the conventional model-based approach. The conventional model-based method for the stripe noise was based on a prior model of the stripe noise and exploited the moment matching approach [56], in which the sensors were assumed to have a linear relationship with each other. Specifically, the conventional model-based algorithm estimates the initial points of vertical stripes and uses edge information to calculate the positions of the noise. The initial points of stripes were calculated based on information of the sensors. Using edge information of the input, homogeneous areas were selected and the start and end points of the noise were calculated based on the initial points of the noise in the homogeneous area. Vertical stripe noise patterns were estimated by subtracting the average value of the areas near the vertical pattern from the average value of the vertical stripe area. In addition, we compared block-matching and 3-D filtering (BM3-D) [57], BayesShrink [58], and weighted nuclear norm minimization (WNNM) [59].

For wave noise removal, multispectral images (RGBN bands) were registered for the wave noise, so we compared our results ($\text{WavCycleGAN}_{\text{RGBN}}$) with various variations to verify the benefits of our framework. Specifically, we generated comparative reconstruction results by the image domain cycleGAN using green channel images (CycleGAN_G), the wavelet directional domain cycleGAN with green channel images (WavCycleGAN_G), and the image domain cycleGAN using multispectral bands ($\text{CycleGAN}_{\text{RGBN}}$). Furthermore, we generated reconstruction results from the GAN model using RGBN bands (GAN_{RGBN}), BM3-D, BayesShrink, and WNNM. We also compared the conventional model-based approach for the wave noise. The conventional method assumed that the panchromatic image can be represented by a linear combination of multispectral band images with the least square regression coefficients [60]. Clean green band images were then calculated using the relationship of the panchromatic image and the multispectral images according to the block-based scheme.

In experiments with real noisy data, we compared our results with all the methods mentioned previously, while our results from the external data were compared with the other five algorithms, with the exception of the conventional model-based approaches which require more information, such as characteristics of the sensors that are not provided in the external data set.

V. Experimental Results

A. Real Experiments

1) Removal of Vertical Stripe Noise: To evaluate the performance of our denoising method for the vertical stripe noise, we visually inspected our results (WavCycleGAN) and compared them with other methods.

Figs. 6 and 7 show denoising results for the image patches from the first scene (agricultural area) and second scene (cloud area), respectively. The reason we chose two drastically different scenes was to validate the generalization capability of our neural network. For Fig. 6, we selected image patches with a size of 400×400 pixels showing significant vertical stripe noise from the first scene. The image patch of size 800×800 pixels was cropped from the second scene for Fig. 7. As shown in the figures, BM3-D and WNNM tended to generate over-smoothed reconstruction results that showed degraded image details and stripes. BayesShirnk failed to remove stripes and produced smoothed edges. GAN removed some stripes while generating severe artifacts. Our results of learning wavelet directional images (WavCycleGAN) effectively removed vertical stripe noise, while results of the image domain cycleGAN (CycleGAN) failed to capture the noise patterns. In particular, our method successfully removed noise without affecting high-frequency components such as edges and textures. Compared with the conventional model-based results, our results demonstrated improved performance in terms of image homogeneity. For instance, in Fig. 7(h), the middle part of the conventional model-based result shows

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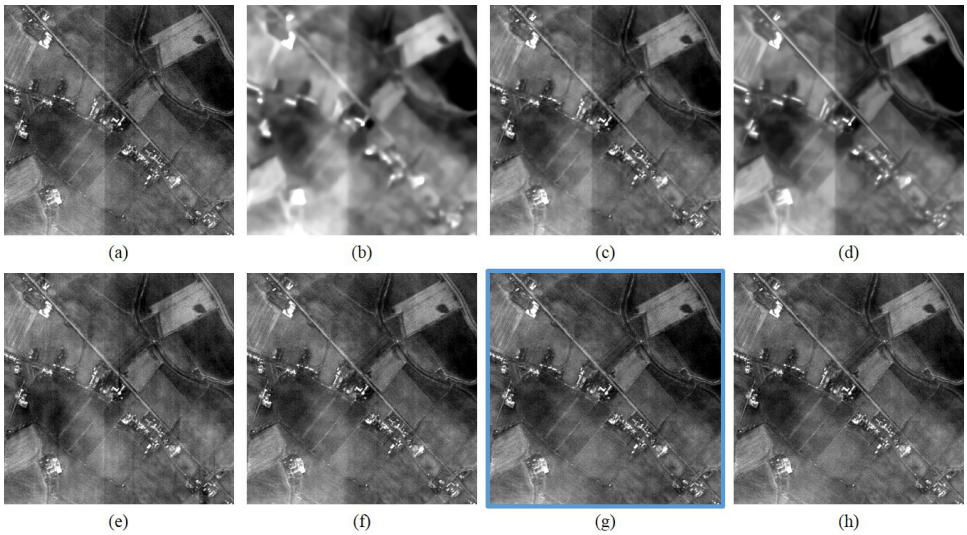


Fig. 6. Results of the vertical stripe noise removal in the first scene (agricultural area). (a) Noisy image, and results of (b) BM3-D, (c) BayesShrink, (d) WNNM, (e) GAN, (f) CycleGAN, (g) our WavCycleGAN, and (h) conventional model-based approach, respectively.

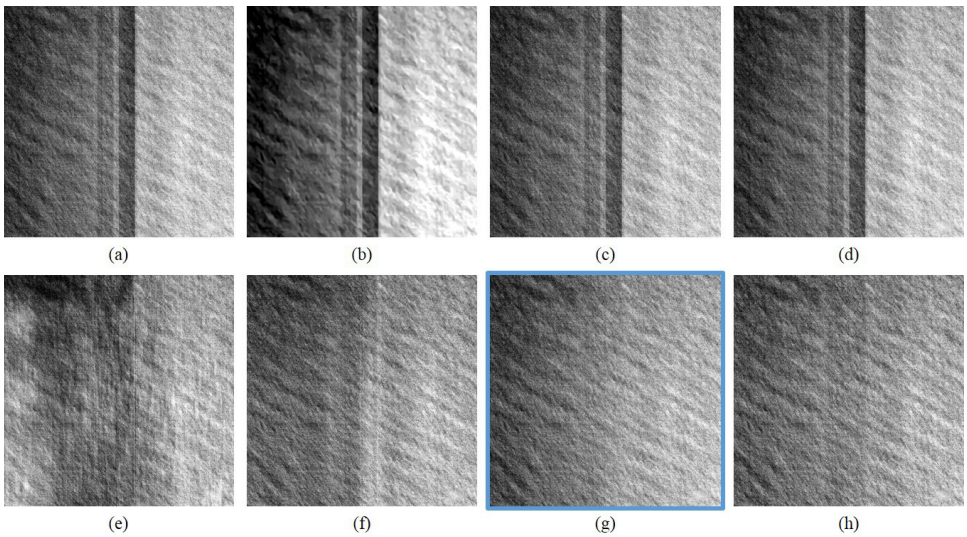


Fig. 7. Results of the vertical stripe noise removal in the second scene (cloud area). (a) Noisy image, and results of (b) BM3-D, (c) BayesShrink, (d) WNNM, (e) GAN, (f) CycleGAN, (g) our WavCycleGAN, and (h) conventional model-based approach, respectively.

image inhomogeneity, while our method shows a homogeneous denoising result.

2) Removal of Wave Noise: Figs. 8 and 9 show results of the wave noise removal in the third scene (ocean) and the fourth scene (cloud), respectively. Again, the reason to show two very different scenes at the test phase is to validate the generalization power of our method. We used the image patch with a size of 200 × 200 pixels for the third scene, and the image patch of a size 400 × 400 pixels for the fourth scene. We also visualized the difference images by subtracting denoised results from the noisy images.

The results of BM3-D, BayesShrink, and WNNM tended to remove both wave noise and image details as shown in Fig. 8. In the fourth scene (Fig. 9), BM3-D, BayesShrink, and WNNM failed to capture wave patterns. As shown in Figs. 8 and 9, the results of the image domain cycleGAN using only the G channel did not successfully remove wave noise. Specifically, we found that CycleGAN$_G$ erroneously removed structural features of objects, while WavCycleGAN$_G$ preserved these high-frequency features. In Fig. 8(f), the difference image contains edges of the object, while only horizontal wave patterns are present in Fig. 8(g).

In addition, our experimental results show that using multispectral images improved performance because the network was able to utilize the spatial correlation between the individual spectral bands for noise reduction. However, results of GAN$_{\text{RGBN}}$ and CycleGAN$_{\text{RGBN}}$ tended to blur the edges and details of images. The difference images of Fig. 8(e) and (h) show that the multispectral image domain GAN (GAN$_{\text{RGBN}}$) and cycleGAN (CycleGAN$_{\text{RGBN}}$) removed high-frequency features, which provide important information for the application of satellite imagery. Compared with

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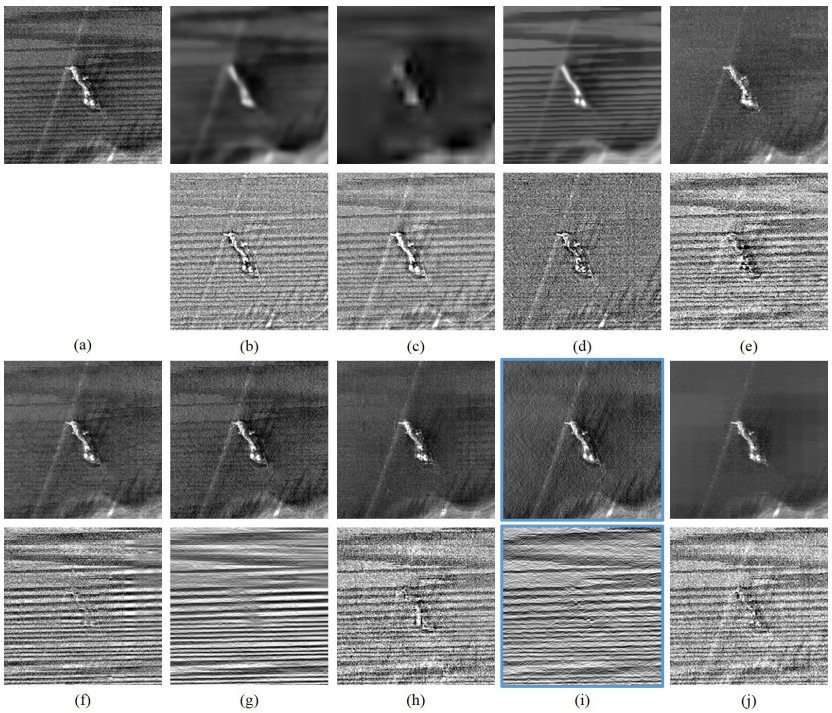


Fig. 8. Results of the wave noise removal (first and third rows) and difference images (second and fourth rows) in the third scene (ocean). (a) Noisy image, and results of (b) BM3-D, (c) BayesShrink, (d) WNNM, (e) GAN_{RGBN}, (f) CycleGAN_{G}, (g) WavCycleGAN_{G}, (h) $\text{CycleGAN}_{\text{RGBN}}$, (i) proposed $\text{WavCycleGAN}_{\text{RGBN}}$, and (j) conventional model-based approach.

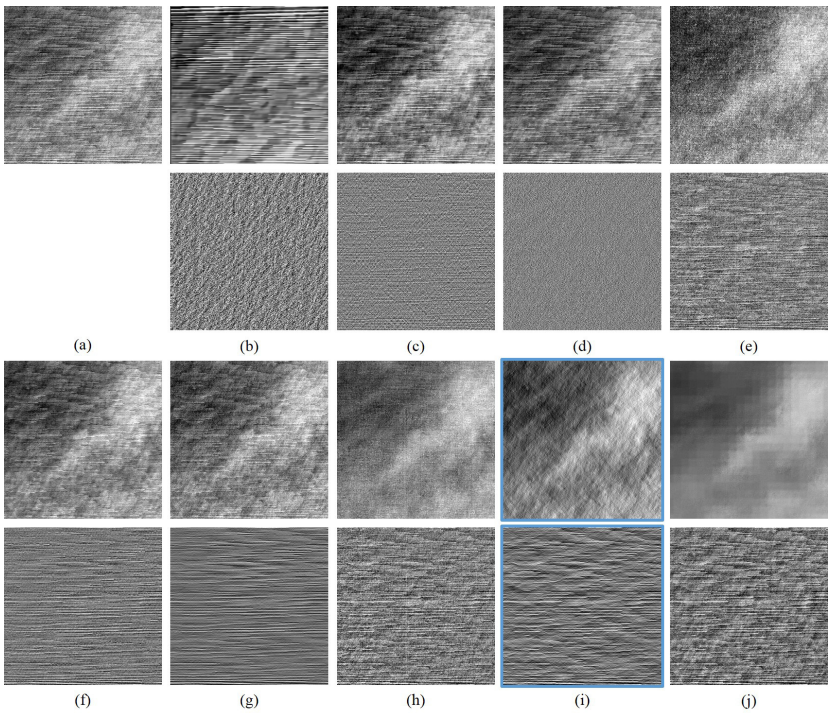


Fig. 9. Results of the wave noise removal (first and third rows) and difference images (second and fourth rows) in the fourth scene (cloud). (a) Noisy image, and results of (b) BM3-D, (c) BayesShrink, (d) WNNM, (e) GAN_{RGBN}, (f) CycleGAN_{G}, (g) WavCycleGAN_{G}, (h) $\text{CycleGAN}_{\text{RGBN}}$, (i) proposed $\text{WavCycleGAN}_{\text{RGBN}}$, and (j) conventional model-based approach.

the results of GAN_{RGBN} and $\text{CycleGAN}_{\text{RGBN}}$, our results using $\text{WavCycleGAN}_{\text{RGBN}}$ confirmed effective removal of noise without sacrificing high-frequency components as shown in Fig. 8(i).

Furthermore, in contrast to our proposed method ($\text{WavCycleGAN}_{\text{RGBN}}$), we found that the use of the multispectral image domain GAN (GAN_{RGBN}) and cycleGAN ($\text{CycleGAN}_{\text{RGBN}}$) often introduced unexpected artifacts to

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TABLE I
QUANTITATIVE COMPARISON FOR THE VERTICAL STRIPE NOISE REMOVAL

Scene #	PSNR [dB] / SSIM							
	Noisy image	BM3D	BayesShrink	WNNM	GAN	CycleGAN	WavCycleGAN	Model-based
1	65.73 / 0.99982	55.19 / 0.99692	58.84 / 0.99878	63.62 / 0.99964	55.79 /0.99842	61.92 / 0.99956	**66.04 / 0.99983**	58.42 / 0.99905
2	66.68 / 0.99985	54.74 / 0.99658	59.86 / 0.99902	64.83 / 0.99973	55.25 /0.99806	64.47 / 0.99976	**67.14 / 0.99987**	61.04 / 0.99946
3	67.02 / 0.99994	61.17 / 0.99932	61.23 / 0.99932	65.66 / 0.99986	55.17 /0.99910	64.60 / 0.99990	**67.93 / 0.99995**	55.12 / 0.99908
4	**66.49** / 0.99994	60.59 / 0.99923	59.96 / 0.99907	64.87 / 0.99983	59.73 /0.99971	63.70 / 0.99988	66.28 / **0.99994**	63.35 / 0.99987
5	63.95 / 0.99992	59.46 / 0.99911	60.04 / 0.99927	63.25 / 0.99984	61.74 /0.99987	63.54 / 0.99991	64.60 / 0.99993	**69.85 / 0.99996**
6	65.00 / 0.99991	59.01 / 0.99887	60.13 / 0.99918	64.36 / 0.99985	64.83 /0.99990	62.62 / 0.99987	65.11 / 0.99991	**68.52 / 0.99993**
7	66.45 / 0.99983	59.25 / 0.99878	59.06 / 0.99877	62.78 / 0.99950	62.20 /0.99960	64.13 / 0.99972	**66.83 / 0.99985**	64.38 / 0.99979
8	63.63 / 0.99971	58.82 / 0.99873	58.96 / 0.99880	61.43 / 0.99938	56.93 /0.99891	61.93 / 0.99958	**65.45 / 0.99981**	64.24 / 0.99974
Average	65.62 / 0.99986	58.53 / 0.99844	59.76 / 0.99903	63.85 / 0.99970	58.95 /0.99920	63.36 / 0.99977	**66.17 / 0.99988**	63.11 / 0.99961

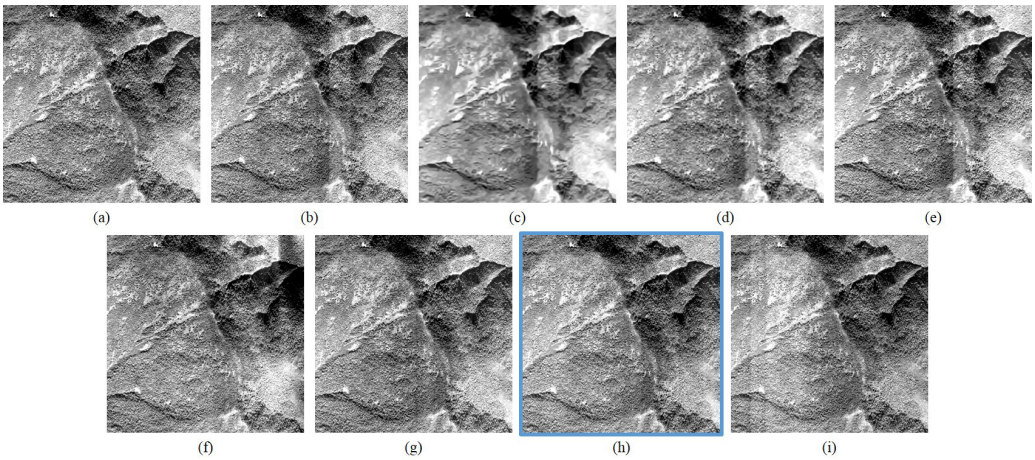


Fig. 10. Results of the synthetic vertical stripe noise removal in scene 8 (mountain area) listed in Table I. (a) Ground truth image, (b) noisy image, and results of (c) BM3-D, (d) BayesShrink, (e) WNNM, (f) GAN, (g) CycleGAN, (h) our WavCycleGAN, and (i) conventional model-based approach.

the green band reconstruction images. For instance, in Fig. 9(e) and (h), the green band reconstruction results are corrupted by other streaks that are not present in the input image. We noticed that these artifacts were from other channel images (R, B, and N bands) during the unsupervised image domain learning. On the other hand, using wavelet directional images with horizontal bands, only horizontal components could be reconstructed, while other directional components could be retained. Therefore, there were no such artifacts in the proposed method as shown in Fig. 9(i).

It is also remarkable that although we used conventional model-based results as a clean domain in training the networks, our unsupervised learning results showed improved denoising performance over the conventional methods by learning the image distribution matching rather than pair-wise matching. For example, the model-based approach completely blurred out the cloud image in Fig. 9(j), whereas our method provided very high resolution image reconstruction without noise in Fig. 9(i).

B. Numerical Simulation

For the quantitative evaluation, we made inferences using synthetic noisy data and calculated quantitative metrics, such as PSNR and SSIM.

1) Removal of Vertical Stripe Noise: Table I lists the PSNR and SSIM values of the scenes from eight noisy images and the reconstruction results by BM3-D, BayesShrink, WNNM, GAN, the image domain cycleGAN (CycleGAN), the wavelet directional domain cycleGAN (WavCycleGAN), and the conventional model-based method (Model-based). The WavCycleGAN outperformed the CycleGAN in terms of PSNR and SSIM for the all scenes. Furthermore, we observed that the mean PSNR and SSIM values of our results were highest among the other methods, which confirms that our method improved the performance by using the wavelet directional image learning scheme. Fig. 10 illustrates results of denoising synthetic vertical noise patterns from the image patch with a size of 600 $\times$ 600 pixels. It is clear that the WavCycleGAN outperformed existing methods, and our method achieved homogeneous image reconstruction while the other approaches did not.

2) Removal of Wave Noise: Table II shows the PSNR and SSIM scores of the scenes from four noisy images and the reconstruction results using the BM3-D, BayesShrink, WNNM, GAN$_G$, image-domain cycleGAN (CycleGAN$_G$), wavelet directional domain cycleGAN (WavCycleGAN$_G$), and results of the conventional model-based method (Model-based). We observed that our method (WavCycleGAN$_G$) outperformed the existing methods in terms of PSNR and SSIM

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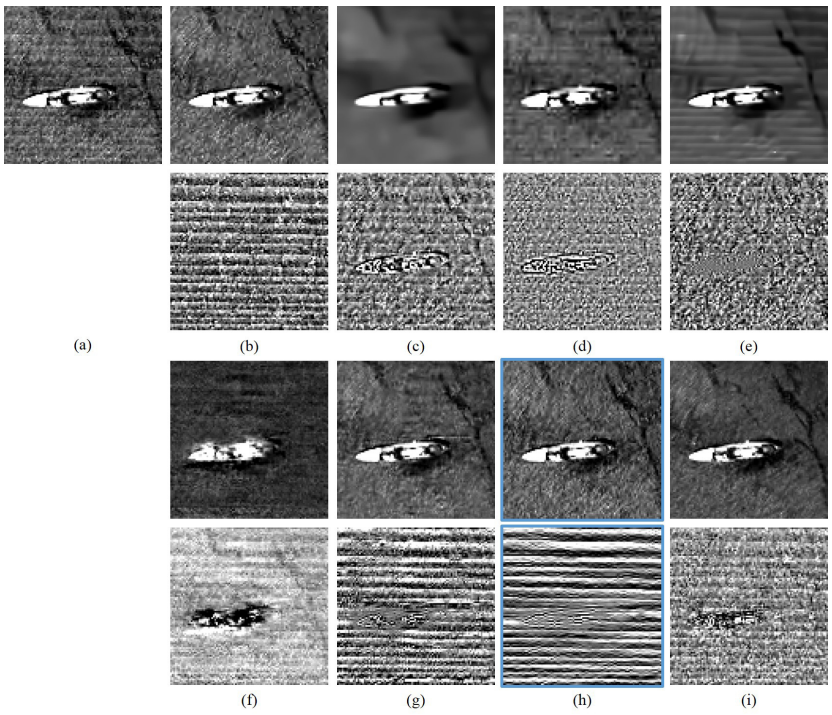


Fig. 11. Results of the synthetic wave noise removal (first and third rows) and difference images (second and fourth rows) in scene 2 (ocean area) listed in Table II. (a) Noisy image, (b) ground truth image, (c) BM3-D, (d) BayesShrink, (e) WNNM, (f) GAN_G, (g) CycleGAN_G, (h) WavCycleGAN_G, and (i) conventional model-based approach.

TABLE II
QUANTITATIVE COMPARISON FOR THE WAVE NOISE REMOVAL

Scene #	PSNR [dB] / SSIM							
	Noisy image	BM3D	BayesShrink	WNNM	GAN_G	CycleGAN_G	WavCycleGAN_G	Model-based
1	51.61 / 0.99275	52.20 / 0.99534	52.83 / 0.99616	52.38 / 0.99537	51.49 / 0.99458	53.41 / 0.99672	52.92 / 0.99467	**57.66 / 0.99836**
2	53.47 / 0.99636	44.47 / 0.97945	42.84 / 0.97921	51.98 / 0.99476	41.42 / 0.97009	53.41 / 0.99678	**53.94 / 0.99686**	49.05 / 0.99348
3	55.94 / 0.99743	47.93 / 0.98688	46.60 / 0.98674	53.34 / 0.99545	50.98 / 0.99380	59.38 / 0.99894	**59.78 / 0.99903**	54.74 / 0.99768
4	62.50 / 0.99948	45.67 / 0.97490	45.88 / 0.97936	51.27 / 0.99249	46.36 / 0.98207	60.01 / 0.99913	**62.63 / 0.99952**	53.25 / 0.99605
Average	55.88 / 0.99651	47.57 / 0.98414	47.04 / 0.98537	52.24 / 0.99452	47.56 / 0.98514	56.55 / **0.99789**	**57.32** / 0.99752	53.67 / 0.99639

values with the exception of Scene 1. Moreover, our method yielded the highest average values of PSNR. Fig. 11 shows the reconstruction results and difference images for the removal of synthetic wave noise using the image patch with a size of 100 × 100 pixels. In contrast to the results of other approaches, our method effectively removed noise patterns without removing edges and detailed information. On the other hand, the other methods removed the details of the scene, resulting in a somewhat blurry image.

C. External Data Validation

In order to evaluate the generalization ability of our model to different data sets, we calculated the average PSNR and SSIM values of the test images from the results of the experiments with UCM data set.

1) Removal of Vertical Stripe Noise: Fig. 12 shows the results of the vertical stripe noise removal for *airplane70* in the UCM data set. For the visualization, we show the image patch with a size of 128 × 128 from the original image. The BM3-D, BayesShrink, and WNNM failed to remove stripes and generated degraded image details. The GAN and CycleGAN produced image artifacts in the reconstruction results. Our WavCycleGAN more successfully removed the noise and maintained high-frequency components of images compared to the GAN and CycleGAN. Table III shows the results of the quantitative comparison for the removal of stripes. Our WavCycleGAN achieved the highest PSNR and SSIM values.

2) Removal of Horizontally Structured Noise: We show quantitative results for the horizontally structured noise removal in Table IV. Our WavCycleGAN achieved the best PSNR and SSIM scores compared to the BM3-D, BayesShrink, WNNM, GAN, and CycleGAN. Fig. 13 shows denoising results and difference images for the horizontally structured noise in *baseballdiamond76* of the UCM data set. As shown in Fig. 13, the BM3-D and WNNM removed edges and details of the image. BayesShrink generated blurred reconstruction results. We observed that the GAN and CycleGAN created severe artifacts, while our WavCycleGAN effectively removed the noise pattern without degrading high-frequency information.

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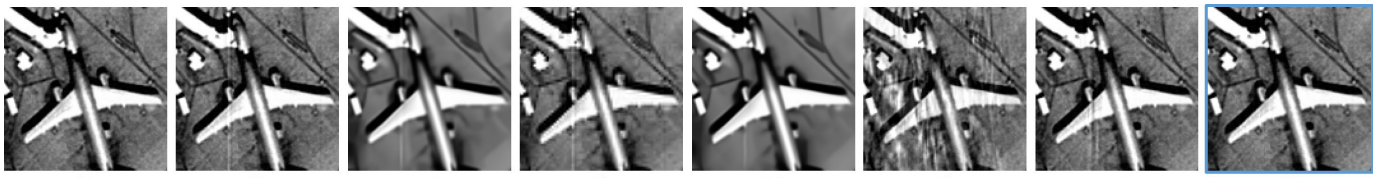


Fig. 12. Results of the vertical stripe noise removal for *airplane70* in the UCM data set. (a) Ground truth image. (b) Noisy image. (c) BM3-D. (d) BayesShrink, (e) WNNM. (f) GAN. (g) CycleGAN. (h) WavCycleGAN.

TABLE III
QUANTITATIVE COMPARISON FOR THE VERTICAL STRIPE NOISE REMOVAL IN THE UCM DATA SET

Index	Methods						
	Noisy image	BM3D	BayesShrink	WNNM	GAN	CycleGAN	WavCycleGAN
PSNR	37.32	29.62	31.17	29.75	24.93	38.37	**40.97**
SSIM	0.96659	0.82017	0.89153	0.82105	0.82942	0.97492	**0.98234**

TABLE IV
QUANTITATIVE COMPARISON FOR THE HORIZONTALLY STRUCTURED NOISE REMOVAL IN THE UCM DATA SET

Index	Methods						
	Noisy image	BM3D	BayesShrink	WNNM	GAN	CycleGAN	WavCycleGAN
PSNR	23.26	24.41	25.68	24.97	25.45	32.15	**32.57**
SSIM	0.58193	0.59210	0.70118	0.60277	0.78097	0.91398	**0.91653**

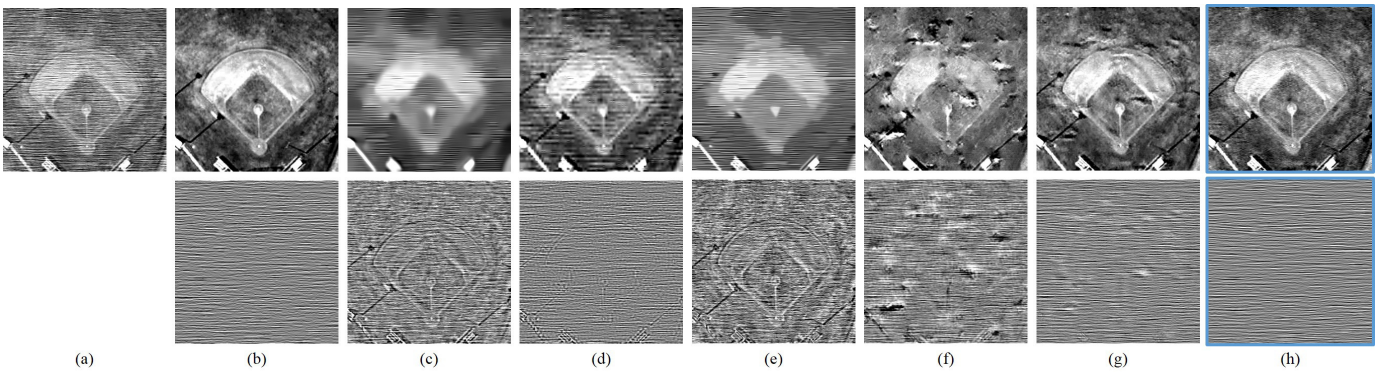


Fig. 13. Results of the horizontally structured noise removal (first row) and difference images (second row) for *baseballdiamond76* in the UCM data set. (a) Noisy image. (b) Ground truth image. (c) BM3-D. (d) BayesShrink. (e) WNNM. (f) GAN. (g) CycleGAN. (h) WavCycleGAN.

Through the external validation with the UCM data set, we confirmed that our WavCycleGAN can be applied to other satellite image sources. In addition, we see that thanks to the use of the wavelet directional image learning scheme, our method effectively reconstructed only noise-related components without affecting edges and image details.

VI. DISCUSSION

A. Ablation Study

To verify the validity of our choice of the 2-D Daubechies-3 wavelet transform (db3), we conducted a parameter analysis and assessed the performance based on PSNR and SSIM values averaged among samples. For the parameter analysis, we trained our model using real noisy data (satellite images) with a varying parameter, while other parameters were set to default values. We evaluated the performance of the models using synthetic noisy data for wave and stripe noise.

1) Wavelet: We used the 2-D Daubechies-3 wavelet transform in our study. To detect fine details of the signal, a small length of a wavelet support is required. The advantage of Daubechies wavelets is that they have a minimal length of a support for a given number of vanishing moments. With larger vanishing moments, the wavelet transform produces sparser representation. Specifically, with sufficiently large vanishing moments, the wavelet transform of the clean signal generates a small number of large coefficients, while the noise signal is represented as small coefficients in the wavelet domain [61]. This is a good starting point for solving a denoising problem because the difference between the clean signal and the noise is well characterized in the wavelet domain. The Daubechies-3 wavelet (db3) has three vanishing moments, which makes the polynomials degree up to two produce zero wavelet coefficients [62].

As an alternative to Daubechies wavelets, Symlets, Coiflets, and biorthogonal wavelets can be used. However, Symlets

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TABLE V
QUANTITATIVE COMPARISON FOR DIFFERENT WAVELETS

Index	Stripe noise removal					Wave noise removal				
	Haar	db2	**db3**	db4	bior3.3	Haar	db2	**db3**	db4	bior3.3
PSNR	65.97	65.83	**66.17**	65.95	65.84	56.37	56.41	**57.32**	56.56	56.59
SSIM	0.99988	0.99987	**0.99988**	0.99988	0.99987	0.99700	0.99721	**0.99752**	0.99725	0.99718

TABLE VI
QUANTITATIVE COMPARISON FOR DIFFERENT WAVELET DECOMPOSITION LEVELS

Index	Stripe noise removal					Wave noise removal				
	Level 7	Level 8	**Level 9**	Level 10	Level 11	Level 4	Level 5	**Level 6**	Level 7	Level 8
PSNR	65.90	66.01	**66.17**	66.01	66.10	55.15	56.52	**57.32**	56.40	56.31
SSIM	0.99988	0.99988	**0.99988**	0.99988	0.99988	0.99691	0.99714	**0.99752**	0.99727	0.99722

TABLE VII
QUANTITATIVE COMPARISON FOR DIFFERENT λ

Index	Stripe noise removal					Wave noise removal				
	0.01	0.1	1	**10**	100	0.01	0.1	1	**10**	100
PSNR	63.76	65.77	65.84	**66.17**	66.05	55.85	55.80	57.27	**57.32**	55.75
SSIM	0.99977	0.99986	0.99987	**0.99988**	0.99988	0.99684	0.99670	**0.99755**	0.99752	0.99650

TABLE VIII
QUANTITATIVE COMPARISON FOR DIFFERENT γ

Index	Stripe noise removal					Wave noise removal				
	0.05	0.5	**5**	50	500	0.05	0.5	**5**	50	500
PSNR	65.57	65.86	66.17	65.99	**66.20**	56.83	57.17	**57.32**	57.19	56.53
SSIM	0.99986	0.99987	**0.99988**	0.99988	0.99988	0.99744	**0.99753**	0.99752	0.99748	0.99719

generate redundant coefficients when the input is real, and Coiflets have longer support than Daubechies wavelets when the same number of vanishing moments is given [63]. Therefore, we conducted the parameter analysis using wavelets such as Haar, Daubechies-2 (db2), Daubechies-3 (db3), Daubechies-4 (db4), and biorthogonal wavelets (bior3.3). A quantitative comparison was conducted using synthetic noisy data and the results are shown in Table V for wave and stripe noise removal. We found that using db3 produces the best averaged PSNR and SSIM.

2) Wavelet Decomposition Level: Determining the wavelet decomposition level is important because it affects the performance of noise removal. With the ideal wavelet decomposition level, the noise signal is presented in certain bands, so that only the wavelet directional image is reconstructed to remove noise components. Table VI shows experimental results for various decomposition levels which confirms that level 9 is suitable for the stripe noise removal, while level 6 achieves the highest PSNR and SSIM values for the wave noise removal.

3) Cycle-Consistency Loss: Cycle-consistency loss is used to mitigate the mode collapsing problem. λ is the parameter to control the ratio of the cycle-consistency loss in the overall loss function (10). If λ is too small, the generators produce artifacts in the output because the one-to-one mapping between the input and the output fails. If λ is too big, the generators try to maintain the input image and then the expressiveness of the generators is reduced. We therefore obtained experimental results for various λ as shown in Table VII. We observed that $\lambda = 10$ gives the best PSNR values in both stripe noise removal and wave noise removal. Using $\lambda = 1$ obtained the best SSIM value in wave noise removal. However, the difference in SSIM values between $\lambda = 1$ and $\lambda = 10$ is small, so we chose $\lambda = 10$ for our experiments.

4) Identity Loss: To induce the identity property, we used the identity loss in our cost function (10). By controlling γ, the ratio of the identity loss in (10) can be changed. Table VIII shows the results of the quantitative comparison for different γ. For both stripe noise and wave noise, using $\gamma = 5$ is a reasonable choice to achieve high performance. For stripe noise removal, $\gamma = 5$ gives the best SSIM value and a comparable PSNR value compared with $\gamma = 500$. For wave noise removal, $\gamma = 5$ achieves the best PSNR value, while the difference in the SSIM value between $\gamma = 0.5$ and $\gamma = 5$ is negligible, so we chose $\gamma = 5$ for our experiments.

VII. CONCLUSION

In this article, we proposed the WavCycleGAN for the multispectral denoising in satellite imagery. The main motivation for using the WavCycleGAN was that our target noise patterns were directionally structured and only directional components of the noise pattern could be reconstructed using the wavelet directional learning scheme for efficient noise removal. Furthermore, to alleviate the problem of unpaired data in practice, we trained the denoising network in an unsupervised manner. Thanks to the use of WavCycleGAN,

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the denoising network could be trained efficiently in an unsupervised manner. Experimental results demonstrated that our method effectively removed noise patterns without sacrificing high-frequency components.

REFERENCES

[1] D. J. Mulla, "Twenty five years of remote sensing in precision agriculture: Key advances and remaining knowledge gaps," *Biosyst. Eng.*, vol. 114, no. 4, pp. 358–371, Apr. 2013.

[2] S. Ø. Larsen, H. Koren, and R. Solberg, "Traffic monitoring using very high resolution satellite imagery," *Photogramm. Eng. Remote Sens.*, vol. 75, no. 7, pp. 859–869, Jul. 2009.

[3] H. M. Pham, Y. Yamaguchi, and T. Q. Bui, "A case study on the relation between city planning and urban growth using remote sensing and spatial metrics," *Landscape Urban Planning*, vol. 100, no. 3, pp. 223–230, Apr. 2011.

[4] S. Voigt, T. Kemper, T. Riedlinger, R. Kiefl, K. Scholte, and H. Mehl, "Satellite image analysis for disaster and crisis-management support," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 6, pp. 1520–1528, Jun. 2007.

[5] Q. Yuan, L. Zhang, and H. Shen, "Hyperspectral image denoising employing a spectral–spatial adaptive total variation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3660–3677, Oct. 2012.

[6] Y. Chang, L. Yan, H. Fang, and C. Luo, "Anisotropic spectral-spatial total variation model for multispectral remote sensing image destriping," *IEEE Trans. Image Process.*, vol. 24, no. 6, pp. 1852–1866, Jun. 2015.

[7] H. Zhang, W. He, L. Zhang, H. Shen, and Q. Yuan, "Hyperspectral image restoration using low-rank matrix recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4729–4743, Aug. 2014.

[8] W. He, H. Zhang, L. Zhang, and H. Shen, "Hyperspectral image denoising via noise-adjusted iterative low-rank matrix approximation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3050–3061, Jun. 2015.

[9] W. He, H. Zhang, L. Zhang, and H. Shen, "Total-variation-regularized low-rank matrix factorization for hyperspectral image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 178–188, Jan. 2016.

[10] B. Du, M. Zhang, L. Zhang, R. Hu, and D. Tao, "PLTD: Patch-based low-rank tensor decomposition for hyperspectral images," *IEEE Trans. Multimedia*, vol. 19, no. 1, pp. 67–79, Jan. 2017.

[11] Z. Wu, Q. Wang, J. Jin, and Y. Shen, "Structure tensor total variation-regularized weighted nuclear norm minimization for hyperspectral image mixed denoising," *Signal Process.*, vol. 131, pp. 202–219, Feb. 2017.

[12] Y. Wang, J. Peng, Q. Zhao, Y. Leung, X.-L. Zhao, and D. Meng, "Hyperspectral image restoration via total variation regularized low-rank tensor decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1227–1243, Apr. 2018.

[13] H. Fan, C. Li, Y. Guo, G. Kuang, and J. Ma, "Spatial–spectral total variation regularized low-rank tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6196–6213, Oct. 2018.

[14] Y. Yuan, X. Zheng, and X. Lu, "Spectral–spatial kernel regularized for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3815–3832, Jul. 2015.

[15] X. Zheng, Y. Yuan, and X. Lu, "Hyperspectral image denoising by fusing the selected related bands," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2596–2609, May 2019.

[16] X. Lu, Y. Yuan, and P. Yan, "Sparse coding for image denoising using spike and slab prior," *Neurocomputing*, vol. 106, pp. 12–20, Apr. 2013.

[17] H. C. Burger, C. J. Schuler, and S. Harmeling, "Image denoising: Can plain neural networks compete with BM3D?" in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 2392–2399.

[18] X. Mao, C. Shen, and Y.-B. Yang, "Image restoration using very deep convolutional encoder-decoder networks with symmetric skip connections," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 2802–2810.

[19] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, Jul. 2017.

[20] K. Zhang, W. Zuo, and L. Zhang, "FFDNet: Toward a fast and flexible solution for CNN-based image denoising," *IEEE Trans. Image Process.*, vol. 27, no. 9, pp. 4608–4622, Sep. 2018.

[21] Q. Yuan, Q. Zhang, J. Li, H. Shen, and L. Zhang, "Hyperspectral image denoising employing a spatial–spectral deep residual convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1205–1218, Feb. 2019.

[22] Y. Chang, L. Yan, and W. Liao, "HSI-DeNet: Hyperspectral image restoration via convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 667–682, Feb. 2018.

[23] Q. Zhang, Q. Yuan, J. Li, X. Liu, H. Shen, and L. Zhang, "Hybrid noise removal in hyperspectral imagery with a spatial–spectral gradient network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7317–7329, Oct. 2019.

[24] J. Guan, R. Lai, and A. Xiong, "Wavelet deep neural network for stripe noise removal," *IEEE Access*, vol. 7, pp. 44544–44554, 2019.

[25] W. Liu and J. Lee, "A 3-D Atrous convolution neural network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5701–5715, Aug. 2019.

[26] W. Shan, P. Liu, L. Mu, C. Cao, and G. He, "Hyperspectral image denoising with dual deep CNN," *IEEE Access*, vol. 7, pp. 171297–171312, 2019.

[27] Y. Chang, M. Chen, L. Yan, X.-L. Zhao, Y. Li, and S. Zhong, "Toward universal stripe removal via wavelet-based deep convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2880–2897, Apr. 2020.

[28] V. Lempitsky, A. Vedaldi, and D. Ulyanov, "Deep image prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9446–9454.

[29] J. Lehtinen *et al.*, "Noise2Noise: Learning image restoration without clean data," in *Proc. 35th Int. Conf. Mach. Learn.*, 2018, pp. 2965–2974.

[30] A. Krull, T.-O. Buchholz, and F. Jug, "Noise2Void-learning denoising from single noisy images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2129–2137.

[31] J. Chen, H. Lin, Y. Shao, and L. Yang, "Oblique striping removal in remote sensing imagery based on wavelet transform," *Int. J. Remote Sens.*, vol. 27, no. 8, pp. 1717–1723, Apr. 2006.

[32] G. Chen and S.-E. Qian, "Denoising of hyperspectral imagery using principal component analysis and wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 3, pp. 973–980, Mar. 2011.

[33] R. Pande-Chhetri and A. Abd-Elrahman, "De-striping hyperspectral imagery using wavelet transform and adaptive frequency domain filtering," *ISPRS J. Photogramm. Remote Sens.*, vol. 66, no. 5, pp. 620–636, Sep. 2011.

[34] E. Kang, J. Min, and J. C. Ye, "A deep convolutional neural network using directional wavelets for low-dose X-ray CT reconstruction," *Med. Phys.*, vol. 44, no. 10, pp. e360–e375, Oct. 2017.

[35] P. Liu, H. Zhang, K. Zhang, L. Lin, and W. Zuo, "Multi-level wavelet-CNN for image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 773–782.

[36] I. Goodfellow *et al.*, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.

[37] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2223–2232.

[38] E. Kang, H. J. Koo, D. H. Yang, J. B. Seo, and J. C. Ye, "Cycle-consistent adversarial denoising network for multiphase coronary CT angiography," *Med. Phys.*, vol. 46, no. 2, pp. 550–562, Feb. 2019.

[39] B. Sim, G. Oh, J. Kim, C. Jung, and J. C. Ye, "Optimal transport driven CycleGAN for unsupervised learning in inverse problems," *SIAM J. Imag. Sci.*, to be published. [Online]. Available: https://arxiv.org/abs/1909.12116

[40] L. I. Rudin, S. Osher, and E. Fatemi, "Nonlinear total variation based noise removal algorithms," *Phys. D, Nonlinear Phenomena*, vol. 60, nos. 1–4, pp. 259–268, Nov. 1992.

[41] H. Huang, R. He, Z. Sun, and T. Tan, "Wavelet domain generative adversarial network for multi-scale face hallucination," *Int. J. Comput. Vis.*, vol. 127, nos. 6–7, pp. 763–784, Jun. 2019.

[42] Y. Liu, Q. Li, and Z. Sun, "Attribute-aware face aging with wavelet-based generative adversarial networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11877–11886.

[43] C. Villani, *Optimal Transport: Old and New*, vol. 338. Springer, 2008.

[44] G. Peyré and M. Cuturi, "Computational optimal transport," *Found. Trends Mach. Learn.*, vol. 11, nos. 5–6, pp. 355–607, Feb. 2019.

[45] X. Mao, Q. Li, H. Xie, R. Y. K. Lau, Z. Wang, and S. P. Smolley, "Least squares generative adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2794–2802.

[46] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[47] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst. (GIS)*, 2010, pp. 270–279.

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[48] J. Fehrenbach, P. Weiss, and C. Lorenzo, "Variational algorithms to remove stationary noise: Applications to microscopy imaging," *IEEE Trans. Image Process.*, vol. 21, no. 10, pp. 4420–4430, Oct. 2012.

[49] Y. Han and J. C. Ye, "Framing U-Net via deep convolutional framelets: Application to sparse-view CT," *IEEE Trans. Med. Imag.*, vol. 37, no. 6, pp. 1418–1429, Jun. 2018.

[50] J. C. Ye, Y. Han, and E. Cha, "Deep convolutional framelets: A general deep learning framework for inverse problems," *SIAM J. Imag. Sci.*, vol. 11, no. 2, pp. 991–1048, Jan. 2018.

[51] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. 32nd Int. Conf. Mach. Learn.*, 2015, pp. 448–456.

[52] D. Ulyanov, A. Vedaldi, and V. Lempitsky, "Instance normalization: The missing ingredient for fast stylization," 2016, *arXiv:1607.08022*. [Online]. Available: http://arxiv.org/abs/1607.08022

[53] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1125–1134.

[54] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*. [Online]. Available: http://arxiv.org/abs/1412.6980

[55] A. Paszke *et al.*, "Pytorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 8024–8035.

[56] F. L. Gadallah, F. Csillag, and E. J. M. Smith, "Destriping multisensor imagery with moment matching," *Int. J. Remote Sens.*, vol. 21, no. 12, pp. 2505–2511, Jan. 2000.

[57] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, "Image denoising by sparse 3-D transform-domain collaborative filtering," *IEEE Trans. Image Process.*, vol. 16, no. 8, pp. 2080–2095, Aug. 2007.

[58] S. G. Chang, B. Yu, and M. Vetterli, "Adaptive wavelet thresholding for image denoising and compression," *IEEE Trans. Image Process.*, vol. 9, no. 9, pp. 1532–1546, Sep. 2000.

[59] S. Gu, L. Zhang, W. Zuo, and X. Feng, "Weighted nuclear norm minimization with application to image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2862–2869.

[60] J. C. Price, "Combining panchromatic and multispectral imagery from dual resolution satellite instruments," *Remote Sens. Environ.*, vol. 21, no. 2, pp. 119–128, Mar. 1987.

[61] S. Poornachandra, "Wavelet-based denoising using subband dependent threshold for ECG signals," *Digit. Signal Process.*, vol. 18, no. 1, pp. 49–55, Jan. 2008.

[62] I. Daubechies and J. C. Lagarias, "Two-scale difference equations II. Local regularity, infinite products of matrices and fractals," *SIAM J. Math. Anal.*, vol. 23, no. 4, pp. 1031–1079, Jul. 1992.

[63] S. Mallat, *A Wavelet Tour of Signal Processing*. Amsterdam, The Netherlands: Elsevier, 1999.



Joonyoung Song received the B.S. degree from the Department of Biomedical Engineering, Yonsei University, Seoul, South Korea, in 2016, and the M.S. degree from the Department of Biomedical Science and Engineering, Gwangju Institute of Science and Technology (GIST), Gwangju, South Korea, in 2018. He is pursuing the Ph.D. degree with the Korea Advanced Institute of Science and Technology (KAIST), Seoul.

His research interests include deep learning for image restoration and image reconstruction in various applications.



Jae-Heon Jeong received the M.S. degree in information communications engineering from Chung-Nam National University, DaeJeon, South Korea, in 2013.

He has been with the Korea Aerospace Research Institute, Seoul, South Korea, since 2014, and has been a Senior Researcher with the Satellite Cal/Val Department since 2014. His primary research interests and back ground are in image processing, computer vision, high performance computing, and automated feature extraction of high resolution satellite image data.



Dae-Soon Park received the B.S. degree in electrical engineering from ChungNam University, Daejeon, South Korea, in 2012, and the M.S. degree in communication system from the ChungNam Institute, Daejeon, in 2014.

He is with the Korea Aerospace Research Institute, Seoul, South Korea. His research interest includes image processing.



Hyun-Ho Kim received the B.S. degree in electrical engineering from Kyunghee University, Suwon, South Korea, in 2016, and the M.S. degree in electrical engineering from the Korea Advanced Institute of Science and Technology, Daejeon, South Korea, in 2018.

He is with the Korea Aerospace Research Institute, Seoul, South Korea. His research interests include image processing, pattern recognition, and deep/machine learning.



Doo-Chun Seo received the Ph.D. degree in civil engineering from Gyeongsang National University, Jinju, South Korea, in 2002.

He has been with the Korea Aerospace Research Institute, Seoul, South Korea, since 2002, and has been a Senior Researcher with Satellite Cal/Val Department since 2005. His primary research interests and back ground are in satellite photogrammetry, sensor modeling, DEM, Ortho-image generation, automated feature extraction from imagery and geometric calibration of high resolution satellite image data.



Jong Chul Ye (Fellow, IEEE) received the B.Sc. and M.Sc. degrees from Seoul National University, Seoul, South Korea, and the Ph.D. degree from Purdue University, West Lafayette, IN, USA.

He was a Senior Researcher with Philips Research, GE Global Research, New York, NY, USA, and a Post-Doctoral Fellow with the University of Illinois at Urbana Champaign, Champaign, IL, USA. He is a Professor with the Department of Bio/Brain Engineering and an Adjunct Professor with the Department of Mathematical Sciences, Korea Advanced Institute of Science and Technology (KAIST), Seoul. His research focus is deep learning theory and algorithms for various imaging reconstruction problems in X-ray CT, MRI, optics, ultrasound, and remote sensing.

Dr. Ye served as an Editorial Board Member for *Magnetic Resonance in Medicine*. He the Chair of the IEEE SPS Computational Imaging TC and the IEEE EMBS Distinguished Lecturer. He was a General Co-Chair of the 2020 IEEE Symposium on Biomedical Imaging (ISBI) (with Mathews Jacob). He has served as an Associate Editor for the IEEE TRANSACTIONS ON IMAGE PROCESSING and the IEEE TRANSACTIONS ON COMPUTATIONAL IMAGING. He is an Associate Editor of the IEEE TRANSACTIONS ON MEDICAL IMAGING and a Senior Editor of the *IEEE Signal Processing Magazine*.

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